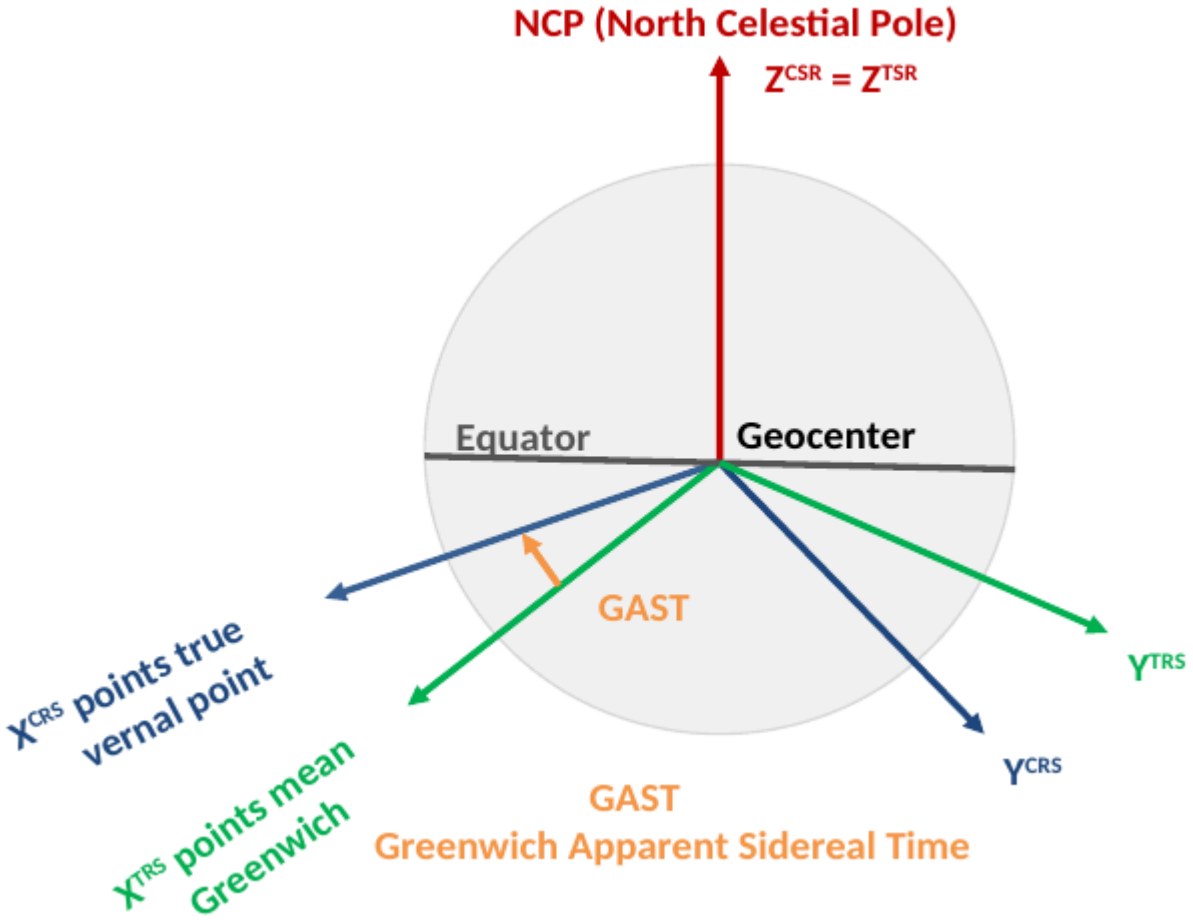
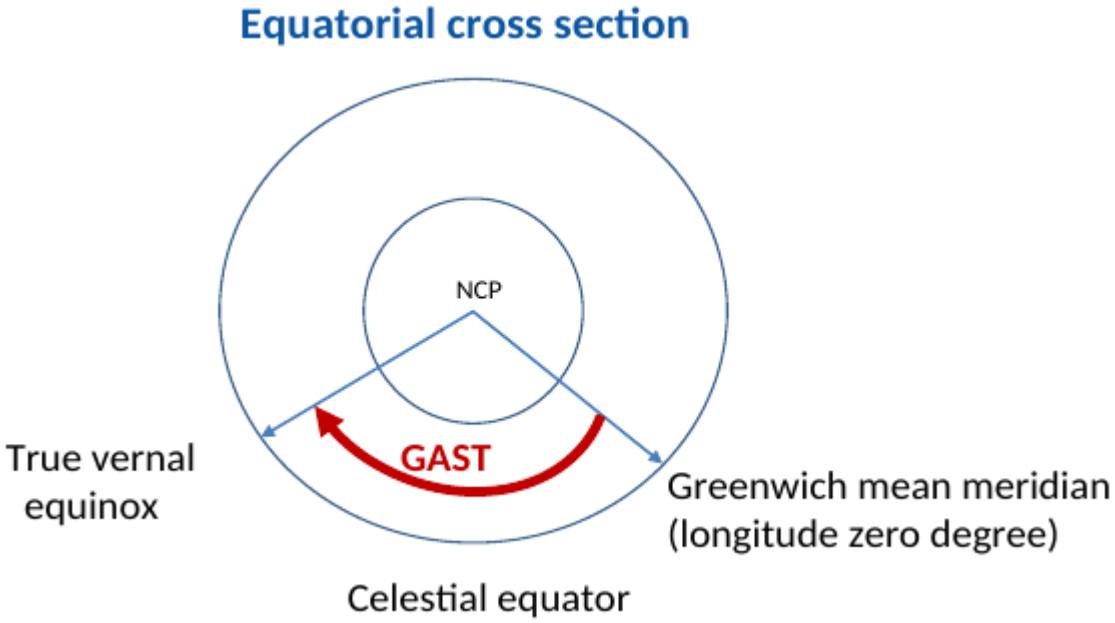
Lecture 3: Representation of Earth Rotation and Earth Orientation Parameters

CRS and TRS Reference Systems



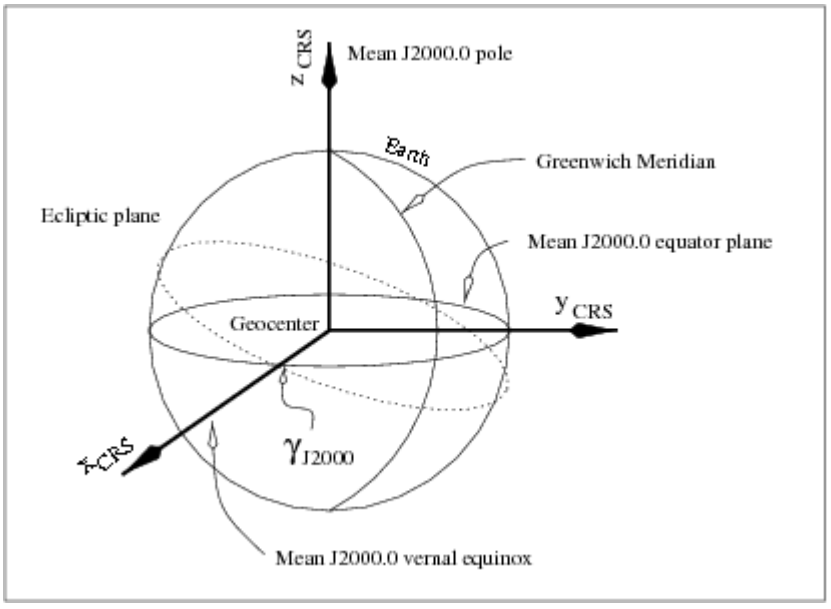
GAST - Greenwich Apparent Sidereal Time

- Key parameter for **transforming CRS and TRS**
  - Mostly tied to time system/definition

- 

  - **Instant** rotation vector of Earth changes **orientation** and **magnitude** due to various geophysical and planetary processes

Conventional Celestial Reference System

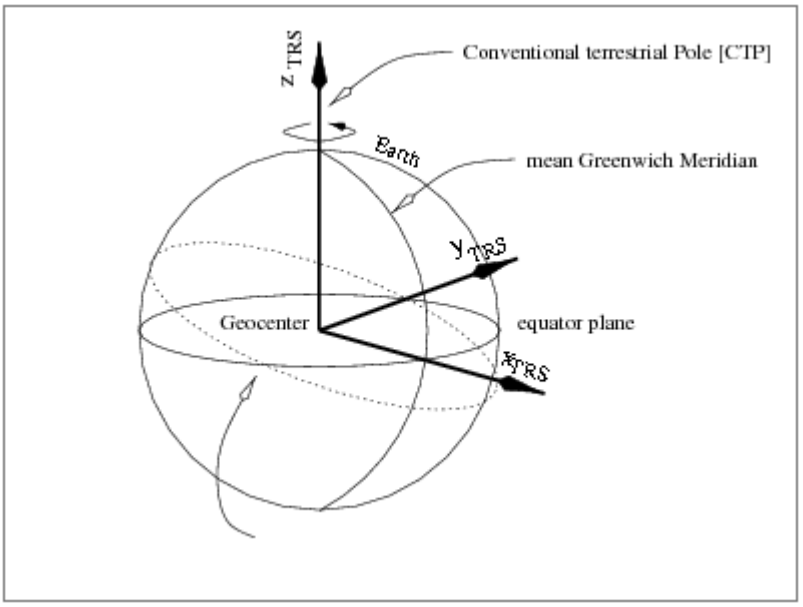


- Z^{CCRS}: direction of the earth **mean rotation** pole at J2000.0*
  - Julian date: number of days passed from noon 01.01.4713 BC
- X^{CCRS}: **Mean** equinox at J2000.0 (intersection between J2000 mean equatorial plane and the ecliptic plane)
- Y^{CCRS}: RHS
- *In CCRS, Earth's instantaneous rotation axis show regular motion relative to mean at J2000.0*

Conventional Terrestrial Reference System

- Z^{CTRS}: Conventional Terrestrial Pole (CTP)

- X^{CTRS}: Intersection between equatorial plane and mean Greenwich meridian
  - Orthogonal to CTP
  - In Mean Greenwich meridian direction
  - Established: International Time Bureau (BIE) observatory and maintained by IERS
- Y^{CTRS}: RHS
- **CTP**(/CIO): average locations of poles from 1900 to 1905
  - In CTRS: Earth's instantaneous rotation axis shows irregular motions relative to CTP

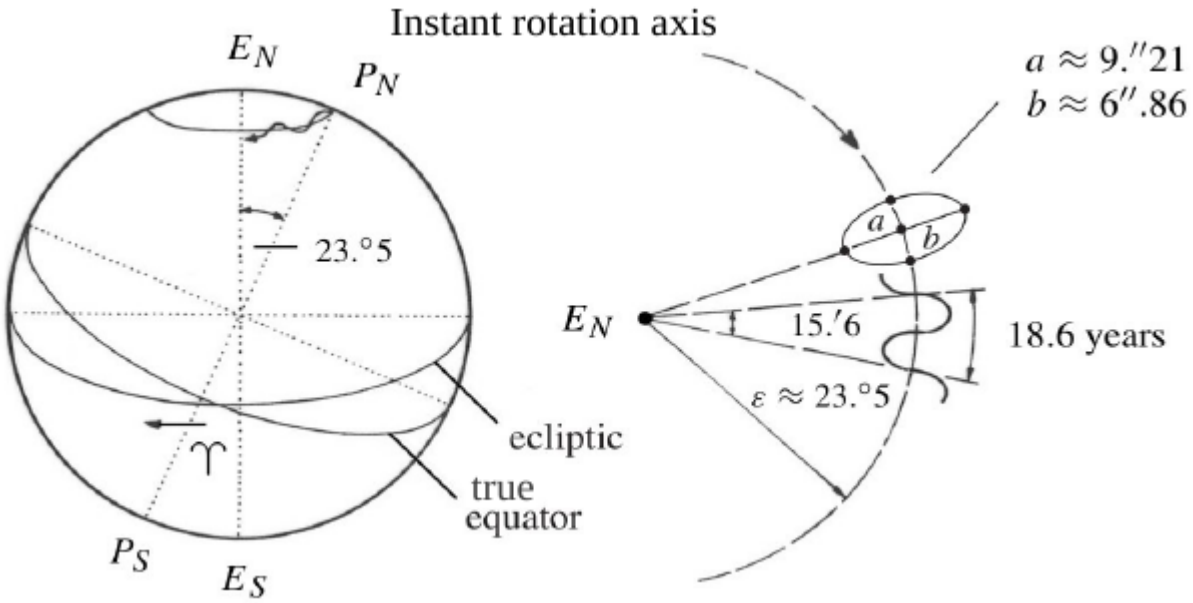


Changes in Rotation orientation in two systems

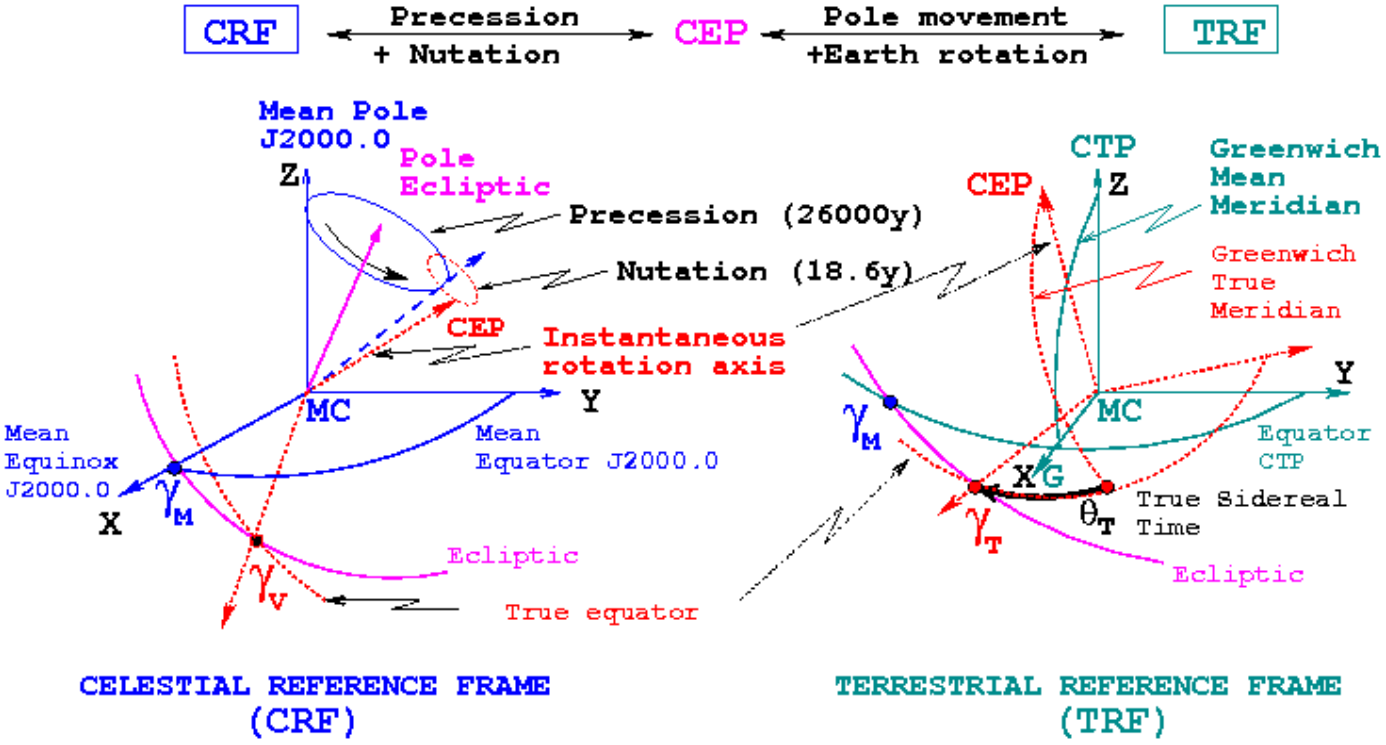
- Instant Earth **rotation axis changes** orientation w.r.t. CCRS *and* CTRS
  - **Angular velocity** fluctutates
    - Important for **time system** and **geophysical modeling** Earth rotation
- Changes in rotation orientation important for **linking two systems**:
  1. Accurate determination of reference frames
  2. Precise navigation and satellite orbits and tranformation between 2 systems
  3. Realisation of time systems

Transforming between CCRS and CTRS

- Precession & Nutation: Temporal variation of the Earth rotation vector in CCRS

- 

  - $P_N P_S$: Earth's instantaneous rotation axis
  - $E_N E_S$: line passing through North and South Ecliptic poles



$$r_{CTRS} = \boldsymbol{SNP} r_{CCRS}$$

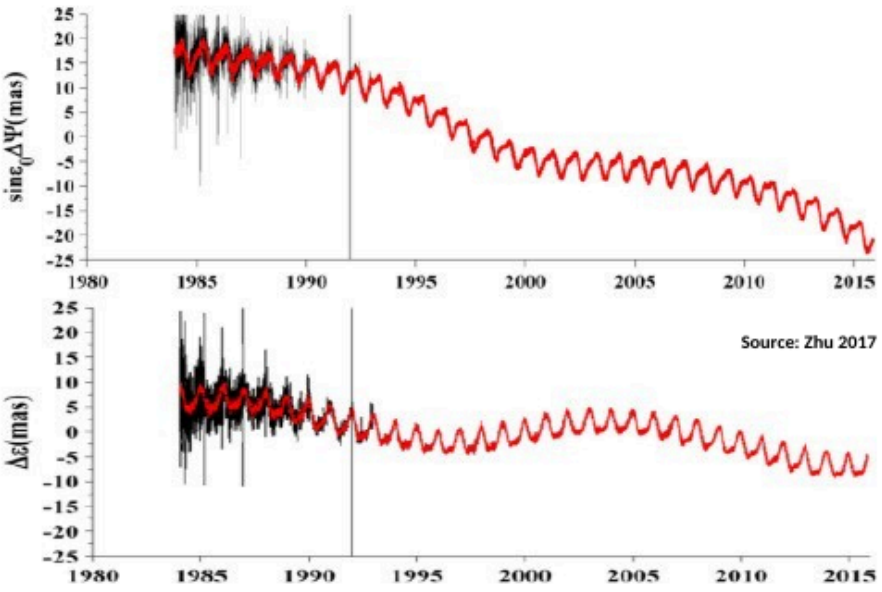
- 3 Rotations for transformation from CCRS to CTRS
  - $\boldsymbol{P}$: Precession
    - Transforms mean equator and mean equinox at the time J2000.0 to **mean equator** and **mean equinox** at time t._
    - secular component (period $26,000a$)
    - $\boldsymbol{P} = \boldsymbol{R}_3(-z)\boldsymbol{R}_2(\vartheta)\boldsymbol{R}_3(-\zeta)$ (with precession angles)
      - Models of precession angles include : IAU 1976, 1980, 2000, 2006

        - z, ϑ, ζ dependent on T, T^2, T^3, with $T = t - t_0$ as the elapsed time from t_0 (J2000.0) from present time t
    - **N**: Nutation
        - Transforms mean equator and mean equinox at time t to **true equator** and **true equinox** at time **t**
        - periodic component (period: $18.6a - 9.3d$(1k different periods))
            - $\boldsymbol{N} = \boldsymbol{R}_1(-\varepsilon - \delta\varepsilon)\boldsymbol{R}_3(-\Delta\Psi)\boldsymbol{R}_1(\varepsilon)$ with
                - $(\Delta\varepsilon, \Delta\Psi)$: the nutation angles
                - Including nutation in obliquity/nutation of longitude
                - ε: obliquity of the ecliptic (instantaneous angle between true equator and ecliptic planes)
                - Values exist e.g.: PREM - spherically symmetric elastic Earth model (IAU 1976, 1980,2000)
                - Nutation observed by space geodetic techniques (VLBI) -> other combinations of data model usable
    - By applying P & N rotations: Transformation of position vector from CCRS(J2000) to true CCRS/CEP (Celestial Ephemeris Pole)
    - **S**: Earth rotation and polar motion
        - $\boldsymbol{S} = \boldsymbol{R}_2(-x_p)\boldsymbol{R}_1(-y_p)\boldsymbol{R}_3(GAST)$
        - (x_p, y_p): coordinates of pole CTRS
        - GAST: Greenwich Apparent Sidereal Times
        - 3 Parameters are: Earth Rotation Parameters (ERP) (In CTRS; N&P: in CCRS)
            - **Cannot be modeled**
            - Computed using observation by international time and latitude service
- ERPs and Precession and Nutation paramter: Earth Orientation Parameers (EOPs) (Variables in Geodesy and Geophysics)

Observing nutation periods and amplitudes

Nutation model IAU 1980

Period	Amplitude			
	Longitude ψ		Obliquity ε	
days	.0001″	.0001″/*cy*	.0001″	.0001″/*cy*
6798.4	-171996.	-174.2	92025.	8.9
3399.2	2062.	0.2	-895.	0.5
1305.5	46.	0.0	-24.	0.0
1095.2	11.	0.0	0.	0.0
1615.7	-3.	0.0	1.	0.0
3232.9	-3.	0.0	0.	0.0
6786.3	-2.	0.0	1.	0.0
943.2	1.	0.0	0.	0.0
182.6	-13187.	-1.6	5736.	-3.1
365.3	1426.	-3.4	54.	-0.1
121.7	-517.	1.2	224.	-0.6
365.2	217.	-0.5	-95.	0.3
177.8	129.	0.1	-70.	0.0
205.9	48.	0.0	1.	0.0
173.3	-22.	0.0	0.	0.0
182.6	17.	-0.1	0.	0.0
386.0	-15.	0.0	9.	0.0
91.3	-16.	0.1	7.	0.0
411.8	-4.	0.0	0.	0.0
117.5	-1.	0.0	0.	0.0
13.7	-2274.	-0.2	977.	-0.5
27.6	712.	0.1	-7.	0.0
13.6	-386.	-0.4	200.	0.0
9.1	-301.	0.0	129.	-0.1
31.8	-158.	0.0	-1.	0.0
...				



Black is VLBI and red is model

Instantaneous (true) CCRS or CEP

- CEP is **true one** (instantaneous earth's rotation axis)
- CEP origin is Earth's centre of mass
  - fundamental plane is defined by true pole -> true equator
- X-axis towards true vernal equinox (intersection between ecleptic plane an true equator plane)
- If external forces absent: CEP: fixed pole in space
- Position of CEP relative to the earth's crust changes due to *irregular mass distribution* of the earth and its variation (*Polar motion*)
- If earth sperical and homogeneous: fixed pole in space

Greenwich Apparent Sideral Time - GAST

- First: Rotation equal to GAST -> transforming true CCRS to mean CTRS
  - 2 other rotations: polar Motion Matrix (mean CTRS -> true CTRS)

$$\boldsymbol{S} = \boldsymbol{R}_2(-x_p)\boldsymbol{R}_1(-y_p)\boldsymbol{R}_3(GAST)$$

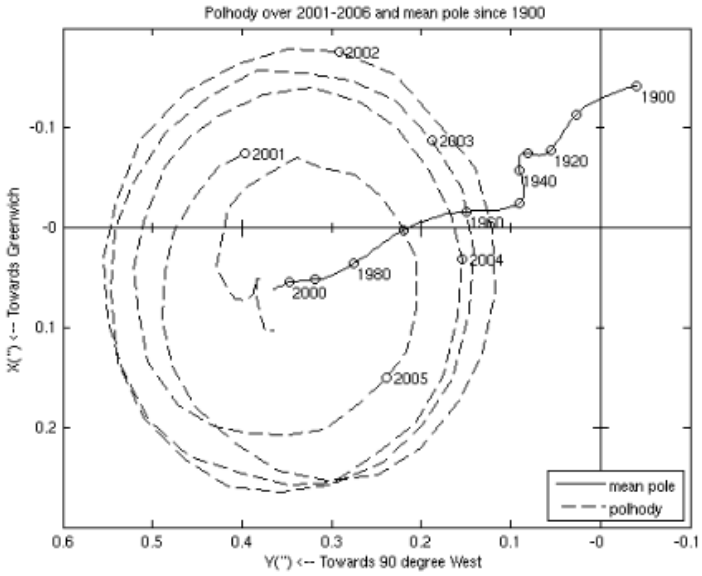
- GAST: Angle between Greenwich mean meridian passing the **true equinox** and represents rotation of Earth around CIP.
- $\boldsymbol{R}_3(GAST)$: diurnal rotation matrix: diurnal rotation around the z-axis of **true CCRS**
- GAST: Expressed as difference dUT=UT1-UT from observations

- Applying GAST rotation: transform true inertial celestial system to Earth-fixed (terrestrial) system

Position vector in mean Earth-fixed system and still doesn't feel variations of Earth rotation due to polar motion.

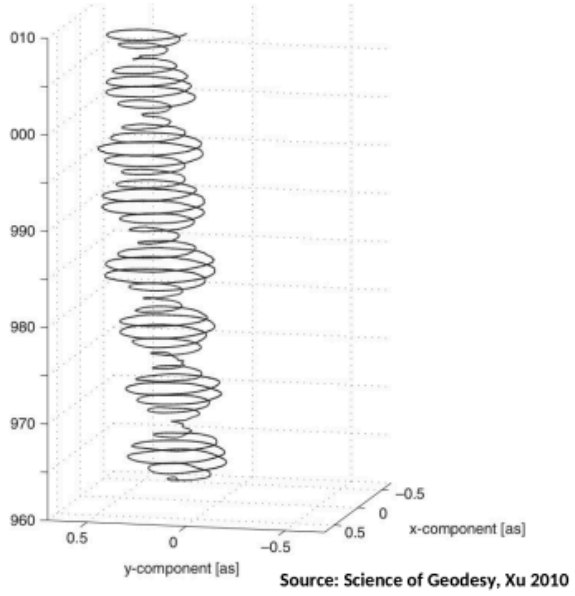
Polar Motion

- True Pole moves w.r.t CTP or CIO
- Polar motion is motion of Earth's rotational axis relative to its crust



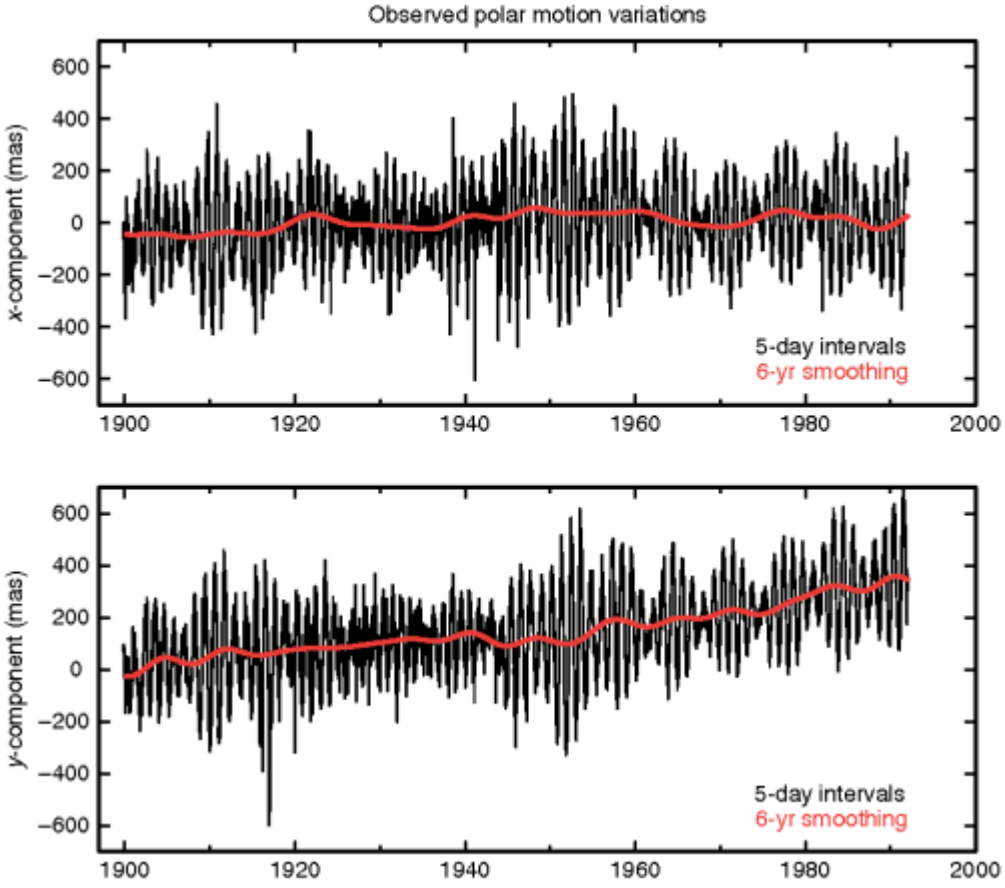
Locations of pole relative to CIO in sec

Source: IERS



Source: Science of Geodesy, Xu 2010

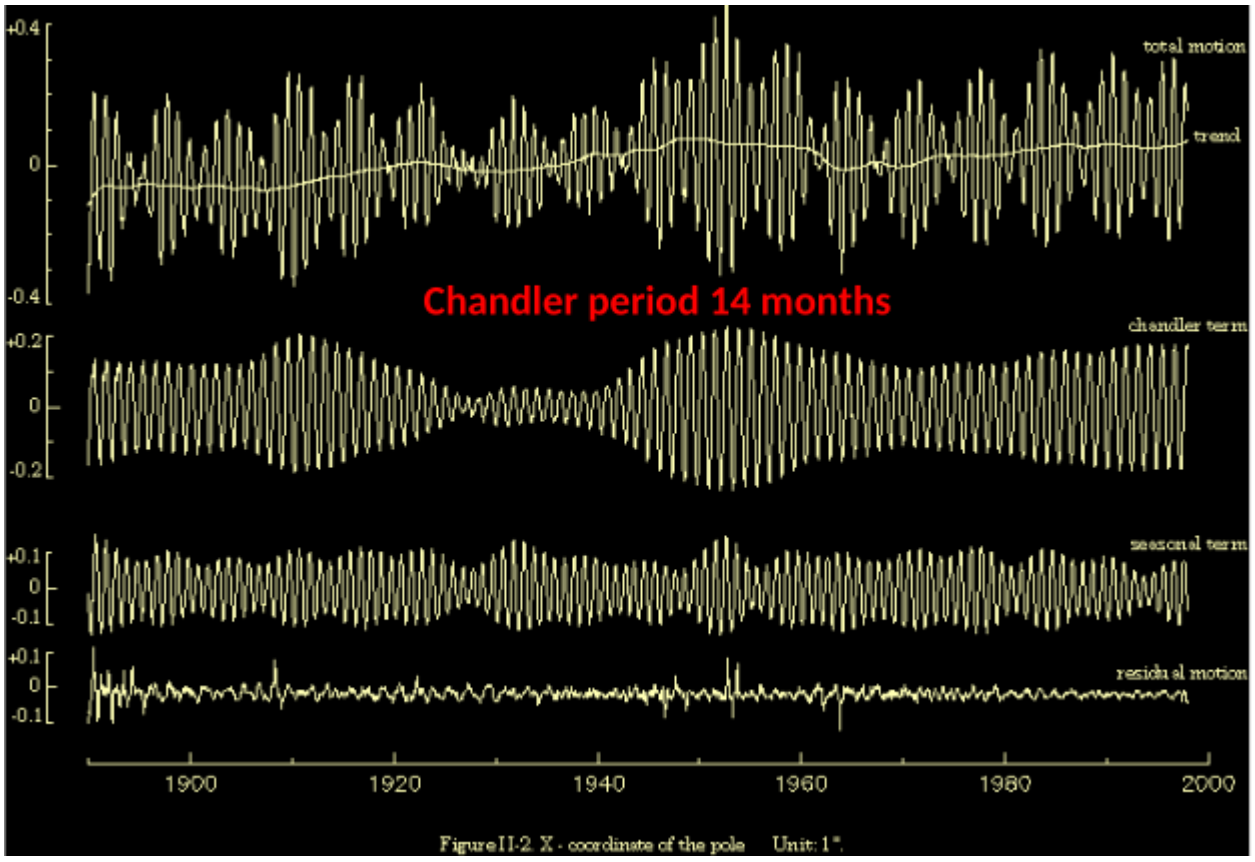
Observations of polar motions from time series of the IERS between 1962 and 2009

- 

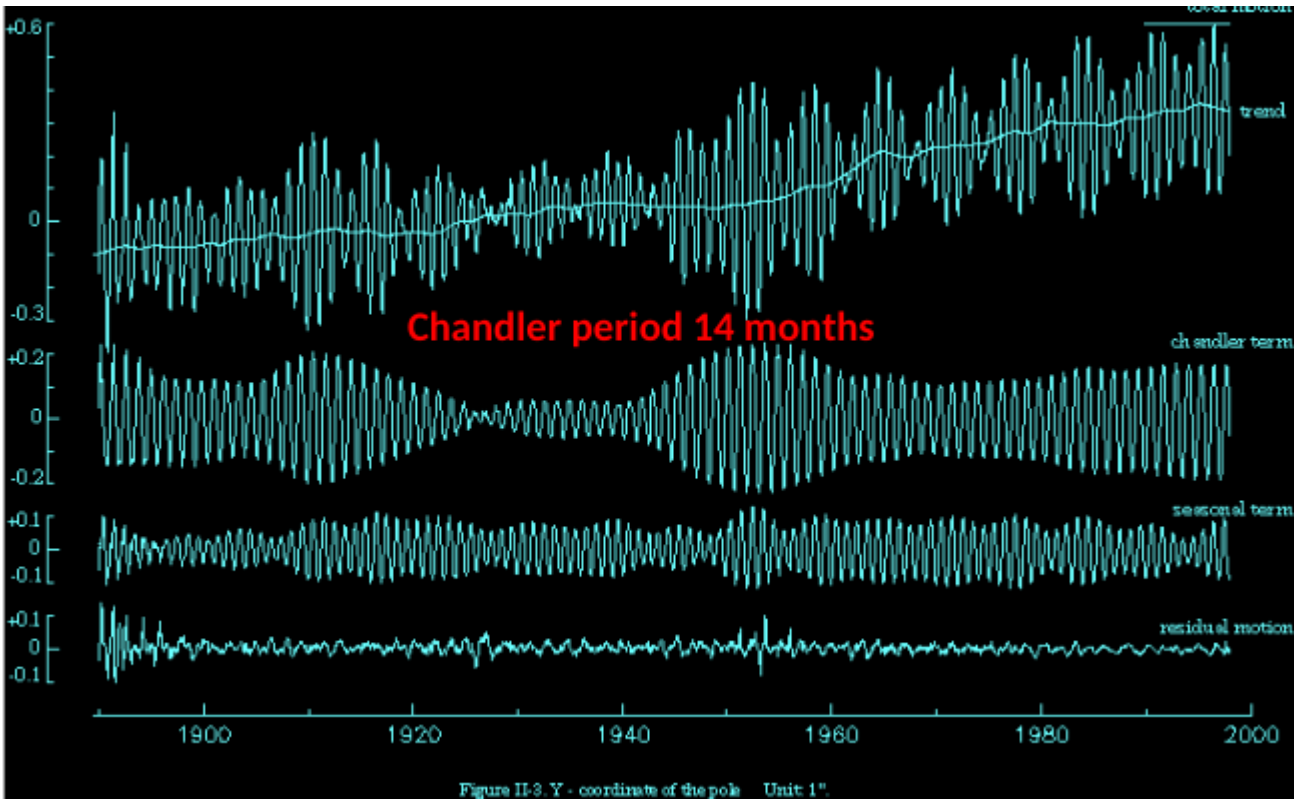
Causes

1. Forced Annual (365d) wobble with amplitude of 100 mas
    - Mostly due to atmospheric, oceanic, hydrologic process (**seasonal excitation**)
2. Chandler wobble (period 433 d) & amplitude 100-200mas (**free wobble**)
3. Quasi-periodic variation on decadal timescale (amplitude: 30 mas)
    - often called: Markowitz wobble + linear trend
4. linear trend (3.5mas/a)
    - Direction towards ~79° W: Due to Glacial Isostatic Adjustment and present-day change in ice sheet mass and glaciers, plate subduction, mantle convection, upwelling mantle plumes, earthquakes
5. Smaller amplitudes - e.g. sub-daily to daily

- X-coordinate of Pole



- Y-coordinate of Pole



From CCRS to CTRS and vice versa

- For coordinate conversion: daily EOPs - based on models and observations
  - e.g. for stars, satellites, observers on Earth

$$r_{CTRS} = \boldsymbol{SNP} r_{CCRS}$$

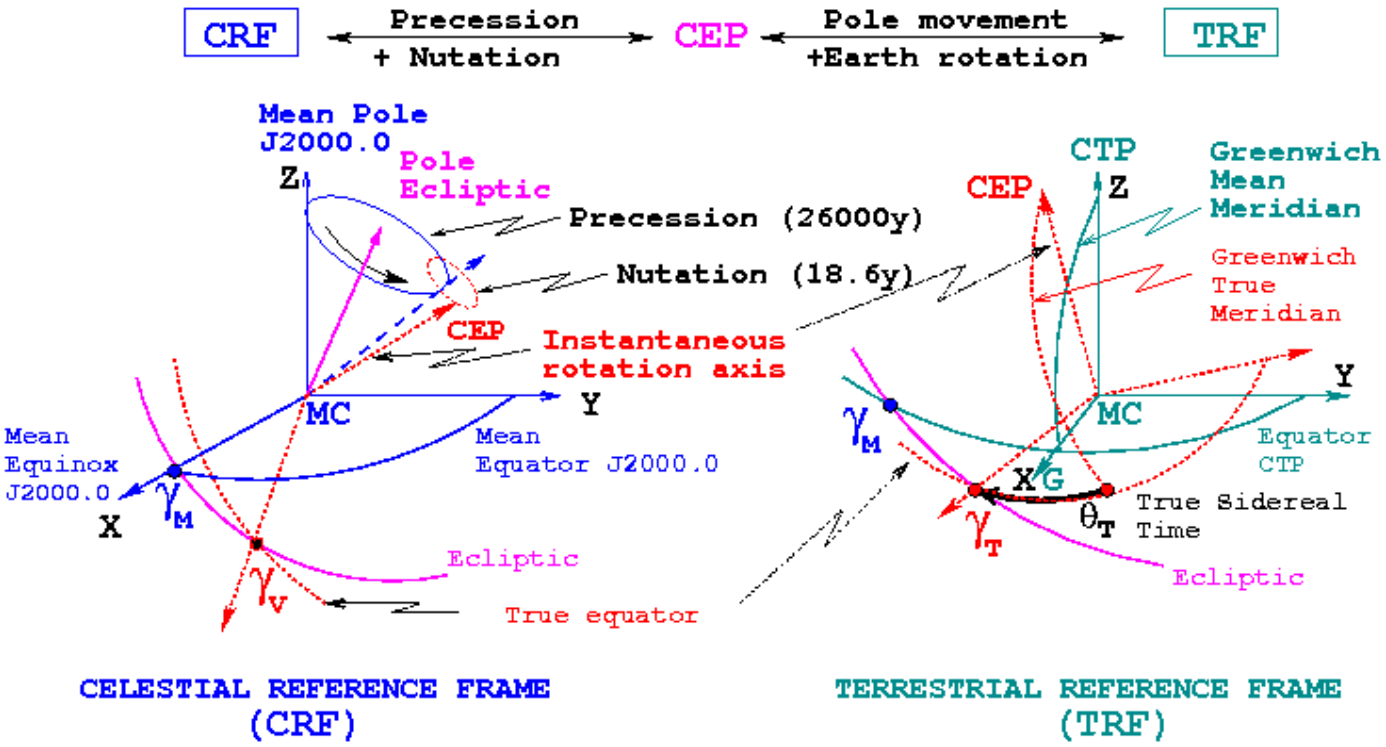
1. Determination of EOPs for maintaining/realising Earth Reference Systems.
2. Monitoring EOPs for understanding geophysical processes and Earth rotation.
3. Using EOPs for transforming 2 reference systems for application includitng GNSS positioning, orbit determination, satellite altimetry, satellite gravimetry etc.

- Data available online

EOP measurement techniques

Technique	Pole positions	UT1	LOD	Nutation parameters
GNSS	✅	❌	❌	❌
DORIS	✅	❌	❌	❌
SLR	✅	❌	❌	❌
VLBI	✅	✅	✅	✅

- Problem: Earth rotation not distinguishable from rotation of satellite orbits
- GNSS provides accurate observations of $\Delta UT1$ and LOD but not absolute values

- VLBI only geodetic technique observing all EOPs

Transforming between CTRS and CCRS